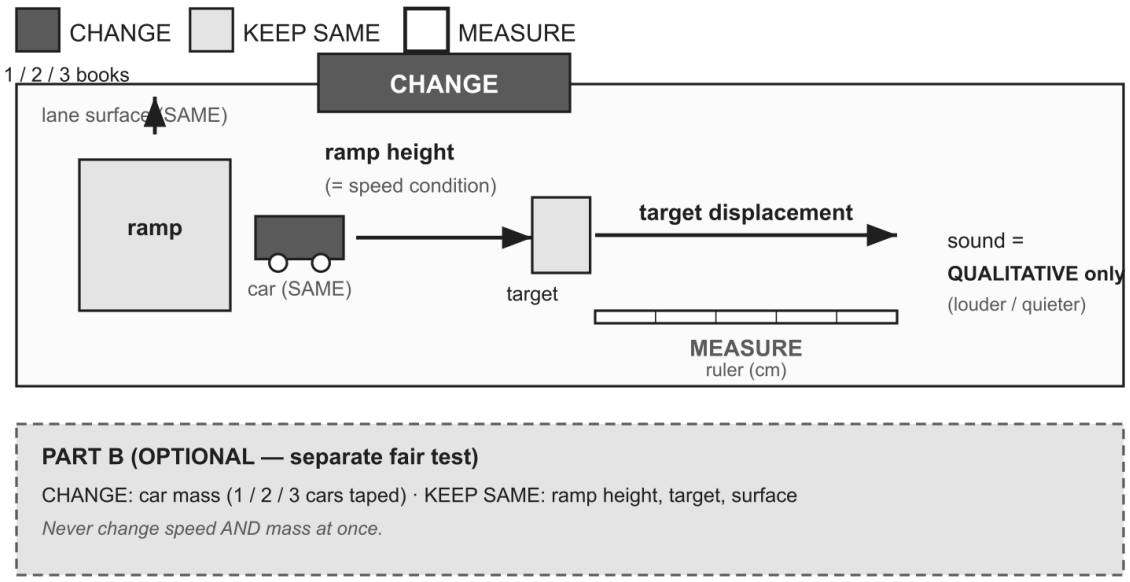
CRASH LAB — SETUP DIAGRAM

Week 3 · Thursday · Teacher setup guide · TEACHER ONLY (laminate for prep binder)



CHANGE	KEEP THE SAME	MEASURE
speed condition / ramp height (1 / 2 / 3 books)	car · target · surface	target displacement (cm)

MATERIALS identical toy car (SAME type) · ramp (cardboard/wood) · books/blocks (3 heights) · target: plastic cup or small block · meter stick / ruler (cm) · tape (start line) · Part B only: 2 extra cars + tape (mass comparison)

PER GROUP 1 car (Part A) · 3 books · 1 target · 1 ruler · tape · (Part B: +2 cars)

SAFETY Clear the crash lane; cars ROLL (never thrown); Newton's cradle = teacher demo only; no heat measurement.

RESET / CLEANUP Return cars/ramps/targets; reset lanes; remove tape.

PREP NOTE Use IDENTICAL cars for Part A. Part B (mass) is OPTIONAL and a SEPARATE fair test — never change speed and mass together.